

1. Determine the following:

- a. Find an integer x such that $0 \leq x < 4$ and $x = 15 \bmod 4$
 $x = 3$. Using, $15 = 3 \cdot 4 + \mathbf{3}$ or $4 \mid 15 - 3$.
- b. $\gcd(2 \cdot 3^{40} \cdot 5^{20} \cdot 13, 2 \cdot 3 \cdot 5^{19} \cdot 7) = 2 \cdot 3 \cdot 5^{19}$
- c. Write the set $A = \{\dots - 3, 0, 3, 6, \dots\}$ in set-builder form.
 $A = \{x \mid x = 3n \text{ for } n \in \mathbb{Z}\}$

2. Complete the following:

- a. $a \cdot s + b \cdot t = 1$ for integers $a, b, s, t \in \mathbb{Z}$. What is $\gcd(a, b)$?
 $\gcd(a, b) = 1$ since $a \cdot s + b \cdot t = 1$
- b. If $a \cdot s + b \cdot t = 1$ is there an integer $d > 1$ such that $d \mid a$ and $d \mid b$? Explain.
No. The greatest common divisor is 1 since $\gcd(a, b) = 1$, and $d > 1$.

3. Complete the following:

- a. Find all the subsets of $A = \{a, b, c\}$.
 $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$
- b. If $|A| = 4$ how many subsets does A have?
 $2^4 = 16$

4. Let $A = \{1, 2\}$ and $B = \{2, 4\}$.

- a. Find $A \cup B = \{1, 2, 4\}$
- b. Find $A - B = \{1\}$
- c. Find $A \times B = \{(1, 2), (1, 4), (2, 2), (2, 4)\}$

5. Prove that $A = A \cup A$.

Proof. ($A \subseteq A \cup A$): Let $x \in A$. $x \in A$ or $x \in A$ so then $x \in A \cup A$. Thus $A \subseteq A \cup A$.

$((A \cup A) \subseteq A)$: Let $x \in A \cup A$. So $x \in A$ or $x \in A$. Thus $A \cup A \subseteq A$. □

6. Let $A \subseteq B$ and $B \subseteq C$. Prove that $A \subseteq C$.

7. Give the definition of the following:

- a. A *1-1 function*.

- b. A *transitive relation*.

8. Let $f : A \rightarrow B$ be a function from set A to set B .

- a. If $|A| = 20$ and $|B| = 21$ can f be onto?

- b.** If f is a constant function and $|B| > 1$, can it be $1-1$? (f is *constant function* if $f(a) = b$ for all $a \in A$ and some $b \in B$.)
- 9.** Let $\mathcal{R} = \{(a, b), (b, a), (b, c), (a, a), (b, b), (c, c)\}$ be a relation on the set $E = \{a, b, c\}$.
- c.** Is $\mathcal{R}$ reflexive? Explain.
- d.** Is $\mathcal{R}$ symmetric? Explain.
- 10.** Define the relation $\mathcal{R}$ on the set of even integers as $x\mathcal{R}y \iff 2|x \cdot y$ (i.e., $x \cdot y$ is an even integer). Show that $\mathcal{R}$ is an equivalence relation.